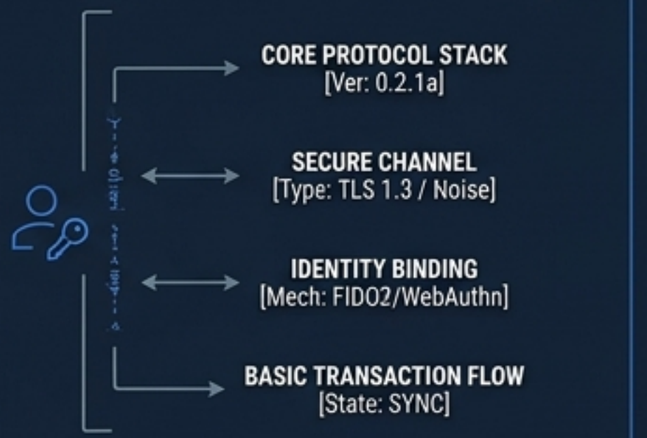
AP2 (Agent Payments Protocol) — Standard Contributed to FIDO Alliance by Google

Built on the AP2 Reference SDK, Prototyping Four Public Roadmap Items

BASELINE VS. V1.x ROADMAP MATRIX

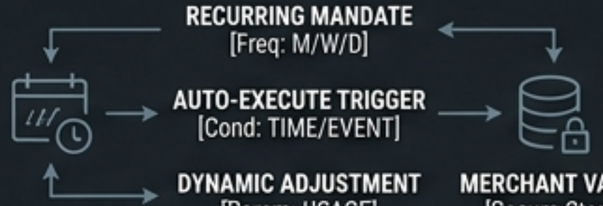
AP2 v0.2 Baseline



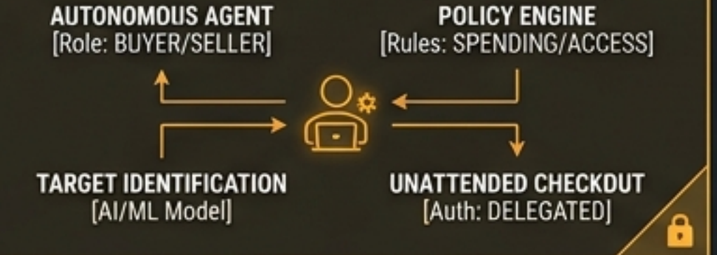
STATUS: STABLE / VERIFIED

V1.x ROADMAP INNOVATIONS

Subscriptions [Process: Auto-Charge]



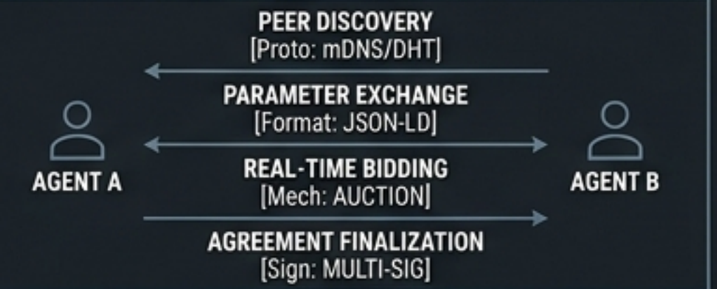
Human-Not-Present [Process: Autonomous Target]



Untrusted Courier Wallet [State: Secure Enclave]



Dynamic Negotiation [Process: Agent-to-Agent]



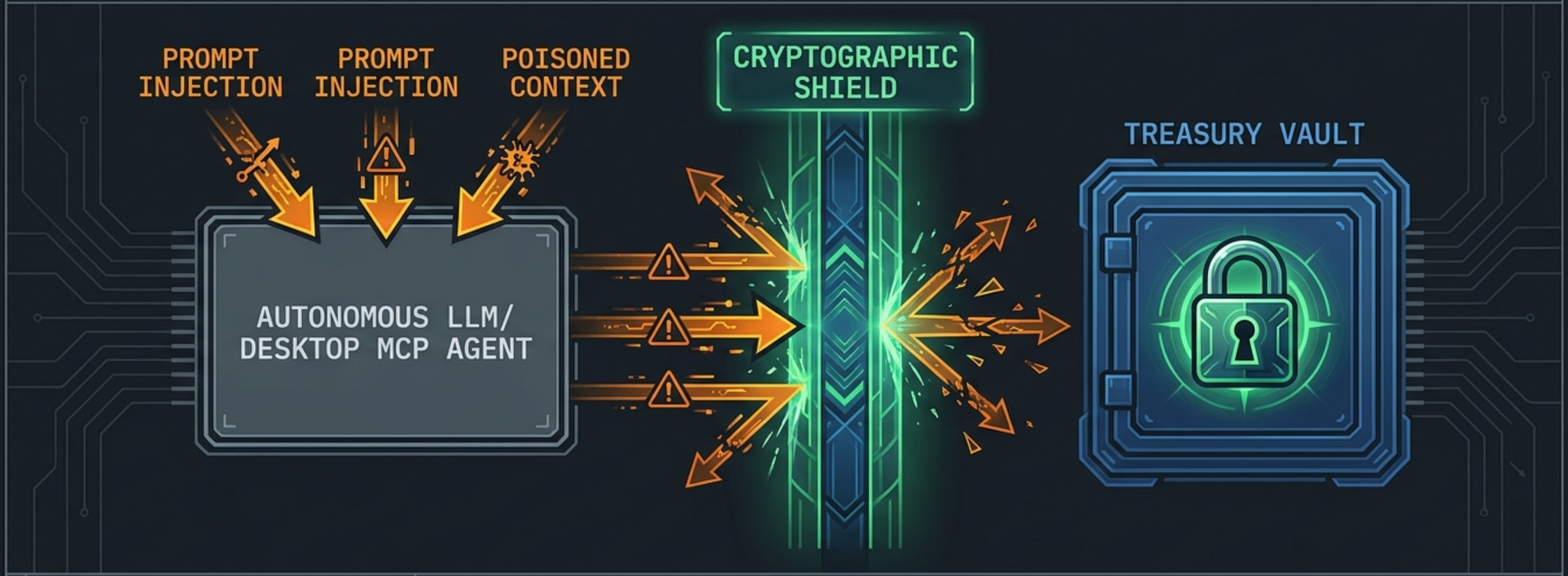
TECHNICAL INVARIANTS

- Full conformance with Google's AP2 v0.2 specification.
- Execution of the complete 3-signer mandate hierarchy:
  1. Owner signs spending limits.
  2. Merchant signs checkout terms.
  3. Buyer Wallet signs final payment mandate.

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THE SECURITY THREAT: WHY LLMs MUST NEVER HOLD PRIVATE KEYS

MITIGATING PROMPT INJECTION AND HALLUCINATION WITH CRYPTOGRAPHIC ISOLATION



TECHNICAL INVARIANTS

- DIRECT LLM KEY ACCESS EXPOSES CORPORATE TREASURIES TO PROMPT INJECTION.
- CONTEXT LEAKAGE ARBITRARILY COMMITS TREASURY FUNDS WITHOUT HUMAN OVERSIGHT.
- ARCHITECTURAL RULE: RELEGATE LLMS TO 'UNTRUSTED COURIERS' HOLDING ZERO KEY MATERIAL.

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TWO-PROCESS FASTMCP ARCHITECTURE

LOCAL LIMIT VERIFICATION BEFORE SIGNATURE GENERATION



TECHNICAL INVARIANTS

- BUYER WALLET HOLDS AGENT KEYS AND OWNER-SIGNED LIMITS LOCALLY (NEVER THE OWNER'S PRIVATE KEY).
- WALLET READS PRICE, TIER, AND MERCHANT STRICTLY FROM THE MERCHANT'S CRYPTOGRAPHIC SIGNATURE, NOT FROM LLM ARGUMENTS.
- UNAUTHORIZED COURIER REQUESTS TRIGGER HARD, LOCAL WALLETREFUSAL BEFORE SIGNATURE GENERATION.

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FRONTIER 1 & 2: RECURRING PAYMENTS & ADVANCE CANCELLATION

BINDING AP2 MANDATES TO PAYPAL VAULT MIT AND SOLVING MULTI-MONTH AUTHORIZATION



TECHNICAL INVARIANTS

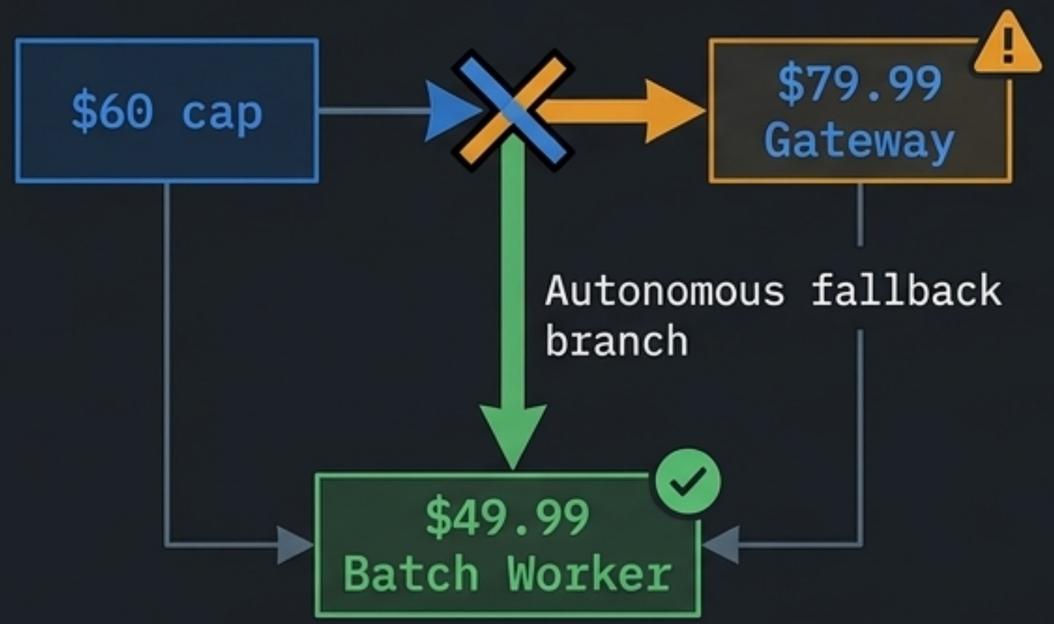
- COUPLES AP2 PAYMENT MANDATES WITH PAYPAL VAULT FOR OFF-SESSION MIT (MERCHANT-INITIATED TRANSACTIONS).
- EACH WEBHOOK PAYLOAD IS CRYPTOGRAPHICALLY RE-VERIFIED AGAINST STORED OWNER SIGNATURES.
- EXECUTES ADVANCE CANCELLATION LOGIC TO INTERCEPT BREACHES BEFORE UNAUTHORIZED FUNDS MOVE.

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FRONTIER 3 & 4: DYNAMIC BARGAINING & ANTI-TAMPER DEFENSE

SUB-BUDGET COUNTER-OFFERS AND ZERO-TOLERANCE 1¢ INFLATION REJECTION

DYNAMIC COUNTER-OFFER LOGIC



ANTI-TAMPER REJECTION TERMINAL

```
{
  "buyer_mandate": "0xFD45A2...",
  "offered_price_usd": 79.99,
  "buyer_cap_usd": 60.00,
  "signature_status": "UNVERIFIED",
  "verification_log": [
    "CHECKING PRICE: $79.99",
    "AGAINST CAP: $60.00",
    "RESULT: EXCEEDS CAP"
  ]
}
                    2026-10-10T14:30:00Z
```

invalid_mandate: BUDGET_CAP_EXCEEDED

TECHNICAL INVARIANTS

- Autonomous multi-tier fallbacks self-heal workflows when buyer caps fall short.
- Strict integer cent math eliminates floating-point rounding exploits.
- ES256 SD-JWT verifier fails closed: altering a single byte or inflating price by 1¢ triggers dual-signed error receipts.

Grounding in Reality: Fleet Operations & Zero-Touch ERP

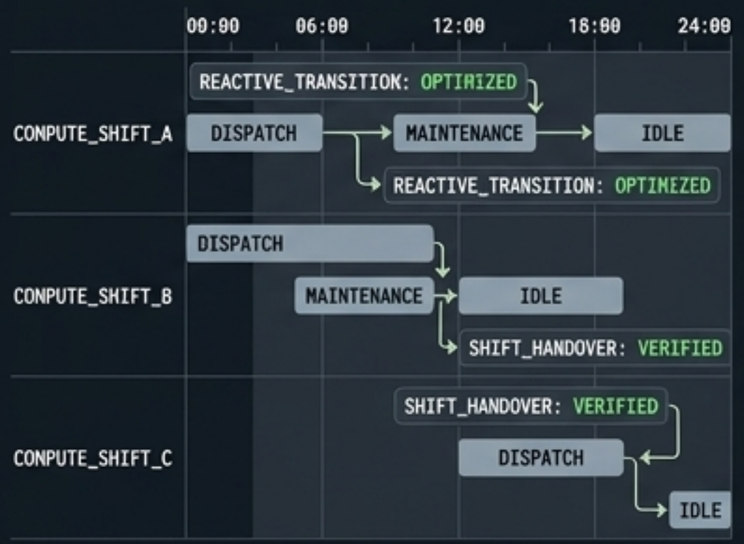
AG Studio v3.0.0, Bryntum 24/7 Dispatch, and Zapier QuickBooks Integration

$14,638.44 Annualized ARR (1463844 integer cents) 2026-10-10

[AG Studio Fleet Ledger]

NODE_ID	STATUS	LOAD_AVG	MEM_USAGE	TRANSACTIONS	BUDGET_BAL
NODE_01	ACTIVE	0.85	84%	146	$1200.00
NODE_02	ACTIVE	0.55	64%	145	$1200.00
NODE_03	ACTIVE	0.55	64%	145	$1300.00
NODE_04	ACTIVE	0.65	67%	92	$1200.00
NODE_05	ACTIVE	0.75	99%	56	1900.00
NODE_06	ACTIVE	6.25	32%	36	$1200.00
NODE_07	ACTIVE	0.55	64%	145	$1200.00
NODE_08	ACTIVE	0.65	52%	103	$1200.00
NODE_09	ACTIVE	0.65	61%	96	$1200.00
NODE_10	ACTIVE	0.85	62%	105	$1300.00
NODE_11	ACTIVE	6.85	50%	56	$1200.00
NODE_12	ACTIVE	0.90	100%	35	$1200.00
NODE_12	ALERTS	0.85	97%	96	$1300.00
NODE_13	ACTIVE	0.85	62%	76	$1200.00
NODE_14	ACTIVE	0.65	82%	105	$1200.00
NODE_15	ACTIVE	0.95	52%	75	$1200.00
NODE_16	ACTIVE	6.75	52%	36	$1200.00
NODE_17	ACTIVE	0.65	59%	26	$1200.00
NODE_18	ACTIVE	0.65	64%	36	$1200.00
NODE_19	ACTIVE	0.65	66%	157	$1200.00
NODE_20	ACTIVE	0.79	32%	87	$1200.00

[Bryntum Scheduler]



[Zapier QBO Exporter]

```
{
  "payload_id": "WEBHOOK_QBO_EXP_789",
  "timestamp": "2026-10-10T14:35:00Z",
  "source": "AG_STUDIO_V3",
  "destination": "QUICKBOOKS_ONLINE",
  "data": {
    "transaction_type": "RECURRING_EXPENSE",
    "amount_usd": 1200.00,
    "gl_account_code": "6300",
    "vendor": "AWS_COMPUTE",
    "status": "EXPORT_READY",
    "verification_hash": "0xAB3F8...D2"
  }
}
```

Technical Invariants:
- Dynamic SQLite WAL hydration reconciling telemetry across 13 active nodes with 0 budget leaks.
- Bryntum Scheduler manages 24/7 compute shifts with reactive in-place visual transitions.
- Zapier MCP automatically routes recurring SaaS expenses to QuickBooks GL Account 6300.

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The Production Gap Report: The Path from Prototype to v1.0

Honest Engineering Analysis of What Production Infrastructure Requires

Hardware Passkeys

Migration from local disk keys to Secure Enclave WebAuthn

Shared State DB

Migration from in-memory nonce caches to distributed Redis/KMS

Mandate Revocation CRLs

Migration from time-based expiration to active Certificate Revocation Lists

Native AP2 Rail Ingestion

Migration from Seller verifier bridges to direct PayPal AP2 rail acceptance

TECHNICAL INVARIANTS
- Prototype keys utilize disk storage; enterprise v1.0 requires hardware-backed passkeys.
- Replay caches require shared database persistence for highly available validation.
- Payment rails must eventually natively ingest and verify AP2 mandates at the network level.

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Empirical Verification & Standards Contribution

557 Passing Tests, 9/9 Judge Audit Checks, and a Battle-Tested Reference Architecture

557 Passed / 9 Skipped (100% Green in ~28s)

(188 reference SDK + 369 project tests)

9/9 Zero-Dependency Judge Checks



Open Source Blueprint

- Untrusted Courier Architecture
- Recurring Mandate Cycles

TECHNICAL INVARIANTS
- 9/9 zero-dependency judge checks execute handshake, bargaining, and provisioning in under 0.2 seconds.
- Strict temp-database isolation guarantees pristine state across 557 test executions.
- Provides a working, empirical reference architecture for the AP2 V1.x working group.